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 Assessment-01 Project Proposal

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Project Proposal

Evaluating the Accuracy, Fairness and Stability of Interpretable Machine Learning for 30-day Hospital Readmission Prediction.

by

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Abstract

The problem of 30-day readmissions after discharge from the hospital is an essential challenge in healthcare that might lead to higher costs of providing care, increased pressure on resources, and even worse results for patients. There have been some positive findings about the application of machine learning algorithms for the prediction of patients at risk of readmission. At the same time, there are several factors, such as calibration, demographic fairness, and interpretability, which should also be taken into account when assessing the potential of a particular machine learning algorithm. In this study, the comparative evaluation of machine learning techniques to predict 30-day readmission will be conducted based on the publicly available Diabetes 130-US Hospitals data set. The performance of the models will be measured by assessing ROC-AUC, PR-AUC, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and calibration using the Brier score and calibration measures. Differences between the performance of the models across different age, gender, and racial groups will be measured through Equal Opportunity and bootstrap confidence intervals. The use of SHAP methodology will help understand the causes behind the model's predictions and explanation stability. Further analysis will be done in order to explore the possibility of decreasing differences in true positive rate through fairness-aware threshold adjustment without decreasing predictive performance and calibration of the models. The goal of this research is to reveal the balance between predictive performance, calibration, fairness, and interpretability of the models, thus determining whether the advanced machine learning models bring any benefits over statistical and clinical methods.

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Chapter 1 Introduction

1.1 Background

Hospital readmission is a key measure of the quality of healthcare delivery, as a sudden return to hospital immediately after discharge often suggests unresolved clinical needs or inappropriate transfer of care, or problems with accessing post-discharge services (Alyaa Abdelhalim PhD, 2024). Thus, predicting readmission risk will help target follow-up treatment and allocation of resources to patients who need them (Raj, 2025). Nevertheless, machine learning models developed for decision-making in healthcare should be evaluated in more detail compared to standard classification methods.

The rise of machine learning is caused by the possibility of detecting complex dependencies between various features such as demographic, clinical, and healthcare utilization features. The importance of traditional statistics models like logistic regression lies in their simplicity, it is possible to see coefficients and interpret results. There are also more complicated models like gradient boosting, random forests and neural networks that can account for non-linear dependencies and interactions, but not all of them.

The rise of explainable AI (XAI) has been partly motivated by this issue. Techniques such as SHapley Additive exPlanation (SHAP) can provide estimates for each particular feature's contribution to a prediction made by the model. The usefulness of SHAP lies in the fact that it allows researchers to obtain both global and individual feature importance in tree-based models. However, the existence of an explanation is no proof that the model itself is interpretable or that the explanations match its logic.

According to Rudin (2019), for cases where it is with crucial importance, researchers need to focus more on building inherently interpretable models rather than explanations of black boxes. This statement is especially true for medicine, since an incorrect explanation might lead to overconfidence in an inappropriate prediction. Collins (2021) argue against the assumption that explainability is always equivalent to increased clinical trust and safety. Thus, the research question in this context is whether it is possible to get appropriate and stable explanations using SHAP.

Fairness presents yet another challenge. It is possible to build a model which works well in terms of overall prediction but operates unevenly for different demographic subgroups. This is especially critical in the case of health care since previous differences in access to health care, diseases prevalence and other factors might be encoded in medical data. Excluding demographic features from the data does not guarantee fairness since other features may act as proxies for demographic features.

There is also an issue of fairness definition. Demographic parity, equal opportunity, equalized odds and calibration stand for different definitions of fair prediction. It is not

usually possible to satisfy all those fairness definitions simultaneously when different populations are characterized by different outcome prevalence rates. Fairness should therefore not be considered as some technical characteristic which could be maximized irrespective of the context. This work thus concentrates on the relation between prediction quality, calibration, fairness and interpretability.

1.2 Research Problem

The study will revolve around three main research questions.

1.2.1 Research Question 01

How well can 30-day readmission be predicted, does a machine-learning model beat plain logistic regression, and which patient factors most influence its predictions?

1.2.2 Research Question 02

Does the model perform equally well for every age, gender and race group, and do the explanations behind its predictions look the same for each group?

1.2.3 Research Question 03

Can standard fairness methods close the gaps, what does that cost in accuracy, and does making the model fairer change what the model says is driving its prediction

1.2 Research Objectives

1.3.1 Main Objective

To examine whether machine learning methods could increase the accuracy of 30-day hospital readmission prediction compared to the statistical method and the clinical benchmark, as well as whether there are differences in predictive performance and fairness of the models among demographic groups.

1.3.2 Specific Objectives

- To develop and compare the penalised logistic regression, interpretable model by nature and XGBoost for predicting 30-day hospital readmission.
- To compare the chosen models with the LACE clinical benchmark in terms of the LACE variables calculable from the dataset.
- To evaluate predictive performance of the models based on discrimination and calibration metrics such as ROC-AUC, PR-AUC, sensitivity, specificity, PPV, NPV, Brier score, and calibration metrics.
- To explore if the predictive performance differs by demographic groups, especially

by age, gender, and race, as well as how the True Positive Rates differ using Equal Opportunity and bootstrapping.

- To explore the predictors for the models and whether the significant predictors are different by demographic groups.

1.3 Scope of the Proposal

The research will concentrate on predicting 30-day readmissions among hospitalized patients with diabetes in the Diabetes 130-US Hospitals dataset. The dataset is publicly available at the UCI Machine Learning Repository and covers clinical services provided from 1999 to 2008 at 130 US hospitals and integrated delivery networks.

The intended cohort of patients will be adults, where individuals under the age of 18 will be excluded from the study. Age less than 18 is a criterion for exclusion as there may be differences between the clinical profile of paediatric and adult patients in terms of clinical characteristics, management, healthcare process, and reasons for re-admission to the hospital. Thus, including paediatric and adult patients in the prediction model together would add more clinical heterogeneity to the model. Nevertheless, considering that age has been provided by the dataset through categories and not precise age, it would not be possible to separate between (0-10) and (10-20). Therefore, the operational cohort definition will be based on the available age categories. Patients whose encounters resulted in deaths or similar hospice-style discharges will be considered for exclusion due to the lack of an opportunity to undergo a conventional 30-day readmission experience. The exact exclusion codes and size of the cohort will be determined during the data audit stage.

The primary outcome measure will be:

1: readmission within 30 days (<30)

0: no readmission within 30 days (NO) or readmission after 30 days (>30)

Some patients have repeated encounters in the dataset. This becomes a critical methodological problem since splitting the encounters randomly might end up with records belonging to the same patient in both the train and test sets. This would result in overfitted results. Therefore, where possible, patient-level division will be considered in the study. The encounter_id and patient_nbr will not be considered as predictive variables.

The main characteristic to consider for fairness in the study is race due to the presence of various races in the dataset. Furthermore, recent literature suggests that fairness should be assessed with respect to racial disparities in healthcare prediction. The secondary characteristic to consider is gender.

1.5 Significance and Purpose of the Proposal

The importance of the proposed study is that it goes beyond the general goal of optimizing predictive accuracy.

Indeed, existing literature shows that the application of machine learning for readmission prediction is possible. For instance, some recent studies have employed XGBoost, LightGBM, and other machine-learning techniques with SHAP-based explanation in order to make predictions on readmissions. Zhang Y (2024), has conducted an analysis of the performance of various machine-learning methods and applied SHAP for the interpretation of the results. Wang HE (2024), has combined machine learning and fairness assessment when predicting readmission, proving that demographic performance disparities must be quantified.

Nevertheless, the presence of such techniques implies that employing only XGBoost, SMOTE, SHAP, and fairness assessment will not be novel enough. The uniqueness of the proposed study thus comes from the comparative analysis and optimization. Such an approach is essential because when assessing a clinical prediction model, it should not only concern its ranking ability. Predicted probabilities have to be reasonably well-calibrated, and there should not be substantial differences in the error patterns for patients of different demographics.

Chapter 2 Literature Review

2.1. Hospital Readmission Prediction using Machine Learning

Classical techniques for prediction of readmission have been using statistical techniques and clinical risk score designs based on human experts. The advantage of machine learning is that it allows one to consider non-linear dependencies between various clinical variables.

Diabetes 130-US Hospitals dataset appears especially suitable for comparative analysis of machine learning because it includes more than 100,000 encounters and many demographic, clinical, administrative and medication variables. According to the UCI repository, it is a classification dataset that has 101,766 observations and 47 variables.

There is plenty of recent evidence supporting the use of machine learning for the task of readmission prediction. For example, Zhang Y (2024) have created explainable machine-learning models for prediction of 30-day readmission in patients with acute heart failure and explained it by SHAP method. Another recent study created an explainable model for 30-day readmission after renal transplantation and used both SHAP and LIME methods for explanation (Alnazari, 2025). Moreover, Emi-Johnson OG (2025) has performed an additional comparison of logistic regression, random forest, XGBoost, and deep neural networks by means of electronic health records for patients with diabetes and revealed that the highest accuracy was achieved by using XGBoost, while according to SHAP, the most important features were previous hospitalizations, drug burden, and comorbidities. Moreover, in the 2025 paper using data from 34 institutions in Nova Scotia, it was shown how explainable ML could not only identify risk factors of 30-day readmissions but also allow for targeted follow-ups after discharge (Maisonnavé M, 2025).

The above evidence highlights the usefulness of explainable machine learning but, at the same time, demonstrates how crowded is the research field. This is why the novelty of this study is not the application of XAI. It is defined by a combination of performance, calibration, fairness and explanation stability comparison with clinical baseline instead.

2.2. Machine Learning Models and Clinical Baselines

A 30-day readmission after discharge from the hospital is considered an important healthcare outcome since it may reflect the problems in patient's recovery process, as well as discharge planning and continuity of care. Early identification of patients who are prone to readmission could help healthcare providers deliver more follow-up and preventive measures to them (Dhaliwal & Dang., 2024). In recent years, there have been more examples that have shown the power of machine learning in predicting patients' readmission in 30 days. For instance, Davis S (2024) investigated machine learning algorithms and survival modelling methods by utilizing longitudinal clinical data and revealed the superiority of readmission prediction performance in comparison to models that used only clinical variables. Likewise, Alnazari (2025) applied an explainable machine learning algorithm for predicting readmission in 30 days in hospitals and showed how explainable artificial intelligence can help interpret risks in clinical applications.

Hence, readmission prediction in hospitals has become one of the key areas of ML application in clinical practice with growing attention to interpretability.

Clinical risk scores have been used for this kind of predictions traditionally. For instance, the LACE index considers such factors as length of stay, admission acuity, presence of comorbidity and emergency department visits before admission as predictors of the probability of readmission (Dreyer, 2026). The HOSPITAL score represents another well-established clinical prediction tool for potentially avoidable readmissions. They serve as good examples due to their clear set of variables and scores. However, due to the simple nature of their structures, they may fail to account for complicated relationships in modern EHRs. Recent studies support this claim, with one done in 2025 comparing LACE score with traditional logistic regression and various machine-learning techniques and finding that the nonlinearity between clinical predictors is likely to negatively affect the predictive capacity of traditional methods (Sharma Y, 2025). Thus, machine learning offers the potential to utilize more EHR variables and detect more complex relationships compared to conventional clinical risk scores.

Another solution offered by machine learning is to learn the association between patient characteristics and outcomes through historical data. Some common machine learning methods used for prediction tasks are logistic regression, decision trees, random forests, support vector machines, gradient boosting approaches, and neural networks. One example is a meta-analysis published in 2023 which compared advanced ML approaches with logistic regression regarding the task of 30-day readmission prediction (Ashna Talwar, 2023). According to the results, machine learning approaches could deliver comparable predictions compared to logistic regression, while the advantage over logistic regression was not always significant in different studies. This finding is critical since it disproves the idea that more complicated models will lead to meaningful improvement in the prediction.

More recent systematic reviews confirm the high interest in ML-based readmission prediction; however, they point out some problems connected with generalizability, methodological quality, and clinical implementation. For instance, Zain Y (2024) aimed at examining ML models predicting 30-day readmission of patients with multiple illnesses shows the high variability in datasets, models, and predictors used, which indicates the dependence of model performance on specific clinical population and data sources. In another recent review of readmission prediction research, researchers have noted the potential for improving generalizability and interpretability of the results (Sharda M, 2025). Additional information on the methodological weaknesses was provided by Selmer N (2025) who discovered that the majority of the analysed readmission prediction studies had a high level of potential bias specifically at the analysis stage, in spite of showing a moderate degree of prediction.

It means that the research question should not be limited to the problem of selecting the most accurate ML model. The clinically meaningful algorithm should also be able to generate valid probabilities, work well with various patient populations and generate interpretable information.

2.3. Fairness in Healthcare Prediction

7 Fairness can be assessed even without altering the model, but it can be done using fairness-aware decision thresholds as well. For instance, a typical approach is for a machine learning model to predict the probability of readmission and then make decisions by using a certain threshold to determine whether the patient belongs to the category of high or low risk.

25
29 For example, the threshold may be set at 0.50 for identifying a patient as having a high probability of being re-admitted. By altering the threshold, the number of patients who will be identified as having a high probability of being admitted varies and hence, alters the true positive and false positive rates. As a result, threshold selection may greatly influence fairness metrics like Equal Opportunity, which emphasizes the importance of equal identification rate of those experiencing the outcome regardless of their demographic group. This relationship has been empirically proven in hospital readmission prediction. Kharrazi (2024) analysed 30-day hospital readmission prediction models for different racial and income subgroups and found large discrepancies in false negative and false positive rates across the populations. Similarly, Wang (2024) showed that threshold alteration in 30-day hospital readmission prediction models results in a compromise between achieving fairness and the overall prediction accuracy. Moreover, a 2025 study used threshold calibration based on subgroup data to address the problem of Equal Opportunity Difference in a real-world readmission prediction and showed that while reducing the discrimination, threshold adjustment influences other factors as well, including accuracy and number of people targeted for intervention (Mackin S, 2025).

39 Recently, there has been empirical support for the idea that threshold selection can result in a trade-off between fairness and overall performance of the algorithm. Specifically, Wang (2024) studied various risk thresholds for hospital readmission algorithms and demonstrated how threshold variations could lead to a trade-off between fairness and overall performance. Similarly, Wang HE W. J. (2024), examined the presence of bias in commonly used 30-day hospital readmission models and raised awareness about the possibility of disparities in the model's performance based on race and socioeconomic status. The study stressed that fairness can only be evaluated through its implications on how risks are used to distribute interventions in health care.

27 But it is important to note that thresholding cannot simply be taken to mean an improvement in the probability calibration of the model because probability calibration involves the actual predicted probabilities and not the thresholding itself. Thresholding could lead to improvement in the classification-based metric of fairness but could alter sensitivity, specificity, and the number of patients chosen for the intervention.

Hence, in the proposed study, it is important to test whether fairness-aware thresholding can lead to improvement in Equal Opportunity without increasing discrimination and probability calibration.

2.4. Explainability and Interpretable Machine Learning

Explainability is yet another problem associated with ML in health care. The use of complex models, for instance, gradient boosting and random forest, can lead to very high predictive power but low direct interpretation of how certain predictions were made. Therefore, such explainability tools as SHAP (SHapley Additive exPlanations) and LIME became quite popular as a solution to the problem.

One such advantage of SHAP is its ability to quantify the importance of each individual feature in a model prediction. For example, SHAP can detect whether the presence of previous hospital admissions for a patient increases their chances of being readmitted, whereas other features reduce their chances. Such an explanation is capable of offering a global view of the factors affecting model behaviour, as well as a local explanation for an individual patient. It has been observed in recent research that explainability can be helpful in identifying any unusual relationship between the model and data (Burns S, 2024). SHAP was likewise used by (Huberts LC (2024) for interpretation of the gradient boosting model for predicting hospital readmission through the use of Shapley values for determining the factors that influence 30-day and 180-day readmissions.

However, one should keep in mind that explainability does not always mean trustworthiness. In their recent paper, Ghassemi, Oakden-Rayner and Beam (2021) state that XAI solutions may provide false confidence in health care and that explanations do not guarantee better decision-making and safety.

In addition, according to Rudin (2019), it is recommended to use inherently interpretable models whenever high-stakes applications allow it rather than using explanations of black-box models. This is especially relevant to the suggested research since the use of penalised logistic regression and other inherently interpretable models can become alternatives to black-box models.

The recent research of hospital readmission has shown that interpretable machine learning models can be constructed for hospital readmission prediction. For instance, Gao X (2023) presented an approach of interpretable modelling which involves extraction of regression tree. It shows that the model can be constructed using the technique of interpretability rather than explaining predictions. Similarly, Ruiz de San Martín R (2024), designed an interpretable machine learning framework for the early prediction of unplanned hospital readmissions. This study showed how both global and local interpretability methods could be applied to find out the factors that play a role in making readmission predictions possible, and thus provided not only the predictions themselves but also their features.

In their research, Lv J (2023) proposed a method that enables explaining predictions by utilizing the machine-learning-based technique for 30-day hospital readmission following stroke. In the research, SHAP was employed to interpret model predictions and identify the most impactful risk factors.

Thus, the suggested research will not just construct SHAP plots and claim the explainability of the model. On the contrary, the study will compare black-box models with the inherently interpretable model and examine the explanation stability. If minor changes in training data lead to major differences in explanations, the confidence in the explanation is lowered.

2.5. Research Gap and Proposed Contribution

From the literature, it is clear that machine learning, explainability, and fairness have individually been studied for their applications in the prediction of hospital readmission. The recent works in this area have focused on the development of explainable machine learning models specifically for diseases like acute heart failure (Zhang Y, 2024), fairness issues related to readmission prediction with special emphasis on race (Wang, 2024), and interpretable machine learning models that perform equally well as complicated algorithms (Gao X, 2023). Hence, it cannot be said that there exists a research gap related to machine learning, SHAP or fairness for the prediction of hospital readmission. Rather, what is required is a comparative study under a controlled environment.

This research project aims at filling this gap by way of empirical comparison between complex machine learning models on one hand and LACE and penalised logistic regression as statistical and clinical baselines on the other hand. More precisely, the objective will be to test whether the use of more complex models results in better predictive performance, whether such improvement in discrimination brings with it an improvement in calibration as well, and whether fairness aware threshold tuning can help decrease discrepancies in true positive rates with the consequence of creating trade-offs in predictive performance or calibration. Also, the stability of explanations under resampling will be investigated to see whether models provide stable explanations, while the performance of an inherently interpretable model will be compared with that of post-hoc explainable machine learning models.

Chapter 3 Research Methodology

3.1 Proposed Approach

The general methodology will involve the following stages:

Stage 1: Dataset Acquisition

The study will employ the Diabetes 130-US Hospitals for Years 1999-2008 data set that is freely available on the UCI Machine Learning Repository website. This data set comprises 101,766 patients' encounters and 47 attributes about inpatient diabetes encounters. The UCI Machine Learning Repository will provide the data for the present study. Prior to analysis, the downloaded data set will undergo an audit process to confirm the number of records, variable names, outcome categories, categorical values, and missing values.

Stage 2: Cohort Definition

The first group to be included in the study shall consist of patients 18 years and above, diabetes inpatients, and records that have a valid readmission status. Unfortunately, the age in the database is recorded using age intervals rather than age itself. Therefore, the (10-20) interval cannot be split into the category of patients younger than 18 and patients between 18-19 years. This constraint shall be noted down and the criteria for the inclusion of patients in the study will be based on the age intervals provided in the database. Encounters of paediatric patients, records with invalid demographics, hospital deaths where 30-day readmissions are impossible, hospice discharge when relevant, duplicated records, and records with an invalid outcome shall be excluded from the study.

Stage 3: Outcome Construction

A transformation of the initial readmission outcome, which is categorized into three classes, will be made to a binary outcome. The encounters that have been marked as "<30 days" in the readmission status will be coded as 1, while encounters marked as ">30 days" or "NO" will be coded as 0. Thus, the main outcome of this study will be the 30-day readmission, which is coded as 1 otherwise 0.

Stage 4: Data Preprocessing

The preprocessing will take into account the handling of missing values, categorical variables, and diagnosis codes without data leakage. As the dataset has a large number of missing values, preprocessing will be fit on the training dataset and then applied to the validation and testing datasets. The numerical features can be imputed with the help of the median imputation method, and the categorical variables can be imputed with an explicit category "Unknown." The

missing value indicator feature can be added for some chosen variables, and the features which have very high missingness can be dropped from the dataset wherever it seems reasonable. The special attention will be paid to weight, payer code, and specialty as they have a lot of missing values. The categorical variables will be encoded in a way suitable for each model, for instance, one-hot encoding will be used for logistic regression and other interpretable models.

Feature engineering techniques will also be applied to construct meaningful predictors such as previous inpatient, emergency, and outpatient utilisation, number of diagnoses, medication burden, treatment changes, age groups, admission characteristics, discharge complexity, and comorbidity-related indicators. All such feature engineering and feature transformations will be built within the training pipeline in order to prevent data leakage from the validation and test datasets.

Stage 5: Patient level data splitting

As some patients have more than one visit in the dataset, it is preferred that the data splitting happens on the patient level. The anticipated data splitting will be 70%, 15%, and 15% for training, validation, and testing respectively, though the percentages may vary based on the final cohort size and the need for stratification. However, the testing set will not be used for selecting the model. This way, there will be no chances of the same patients appearing in both the training and testing sets.

Stage 6: Model Development

Overall, three models will be developed to create a balanced comparison between predictive performance and interpretability. The first model to be developed will be penalised **logistic regression**, which will be employed as a statistical **baseline**, since it is a widely-used and fairly interpretable method of clinical prediction. It is going to serve as a basis for determining whether there is any added value from more complex machine learning models. Hyperparameter tuning will be performed using cross-validation.

The second model to be developed will be Explainable Boosting Machine (EBM). EBM will be employed because of **its ability to detect non-linear relationships and interactions** while being an inherently interpretable model at the same time. It is going to help examine whether there is any improvement in predictions using non-linear modelling without resorting to very complex black-box model. The third model to be developed will be XGBoost, which will be considered a high-performing nonlinear machine learning model.

Other **machine learning algorithms, including random forests, support vector machines, neural networks, and deep learning models**, will not be covered because including too many algorithms in the research will raise its complexity and computational load without helping to achieve the research goals. The three chosen models give different levels of algorithmic complexity: an ordinary statistical model, a built-in interpretable nonlinear model, and a nonlinear model with high performance. Thus, it will be possible to analyse how increased complexity brings better performance and interpretation.

Stage 7: LACE Baseline

2 The LACE score will be implemented only where it is possible to reconstruct all the elements needed from the dataset. The results will be analysed using ROC-AUC, PR-AUC, sensitivity, specificity, calibration, and Brier Score. It will be made explicit which elements of LACE are not possible to be reconstructed from the dataset. The LACE score will also be used as the alternative measure for clinical purposes since it was specially designed for predicting mortality or readmissions.

Stage 8: Calibration

6 Calibration will be very critical in the entire process of evaluation. For instance, when the model estimates that each of the 100 people has a 20 percent chance of being readmitted, roughly 20 percent of those people ought to have that experience in reality. As such, this process of evaluation will entail the use of calibration curves, Brier score, calibration intercept, and calibration slope.

Stage 9: Explainability

SHAP will be primarily applied to interpret the predictions of the XGBoost model. The feature explainability analysis will include global feature importance, SHAP summary plots, and individual patient explanations. Additionally, dependence plots will be applied wherever necessary to analyse the relationship between the key features and model predictions.

3.2 Expected Outcomes/Results and Limitations

Outcome 1: Comparative model performance

This research aims at discovering whether XGBoost, logistic regression, or the inherently interpretable model delivers the highest prediction performance. However, it cannot be assumed that XGBoost wins the race since a good performance by the logistic regression model will help to reinforce the argument regarding the preference for simplicity in an important context.

Outcome 2: Clinical baseline comparison

34 This research will show whether the machine-learning model performs better than the LACE index. It would be an important finding in case the machine-learning model does not perform better than LACE.

Outcome 3: Calibration evidence

This research will reveal whether the best-discriminating model is also well-calibrated. A model with a slightly smaller AUC may still have a better-calibrated risk estimate.

Outcome 4: Fairness finding

The investigation will estimate the presence of group disparities in the true-positive rate of the model. It may be that the unmitigated model would display different sensitivity by race. Then, fairness-aware adjustment of the threshold will be performed.

Outcome 5: Fairness-performance trade-off

The research will reveal whether enhancement of equality leads to:

- decrease in AUC
- decrease in PR-AUC
- worse calibration
- increase in false-positive rates

In itself, this trade-off will constitute an important research discovery.

Outcome 6: Explanation stability

It is anticipated that the research will detect factors that predictably affect model predictions. Differences in SHAP rankings across bootstrap samples will undermine the hypothesis of stability of individual explanation.

Outcome 7: Evidence-based model recommendation

The recommended model is not required to have the best AUC score. The choice of the preferred model will depend on such criteria as:

- discrimination
- calibration
- fairness
- interpretability
- stability of explanation
- threshold performance in clinic

3.2.2 Limitations

Following table lists the few important limitations

Table 3.2: List of expected limitations		
S. No.	Limitations	Description
01	Dataset Generalisability	Since the study will use data collected in a particular healthcare institution, then the results might not necessarily generalise to other hospitals, patient populations, or other countries' healthcare systems.
02	Clinical Benchmark Availability	The LACE Index will only be considered where the necessary variables for calculating it can be derived from the available dataset. The presence of missing values might affect the accuracy of the clinical comparisons made.
03	Historical and Observational Data	Since the dataset is from the historical patient records, the study can only establish an association between patient characteristics and readmissions but cannot establish causation
04	Fairness Metric Scope	The concept of fairness being considered in this project is Equal Opportunity. This is just one way of defining algorithmic fairness.
05	Explainability Evaluation	There might not be sufficient time in the project timeline to conduct evaluations of explanations by healthcare professionals due to the project deadline. Therefore, the explanation quality will be computed computationally through explanation stability.
06	Project Time Constraint	The research must be conducted in the available 18-week time frame.

3.3. Ethical Consideration

Machine learning in the field of healthcare carries certain ethical, legal, and cultural implications since predictions will affect decisions concerning patients.

Regarding the research under discussion, it will rely on a public and de-identified dataset, which means that this research will not have access to any personal data of patients. Still, there will be certain ethical concerns. Regarding privacy and confidentiality, any attempts at identifying people included in the dataset will not be made. The data will be stored securely and used only for research purposes.

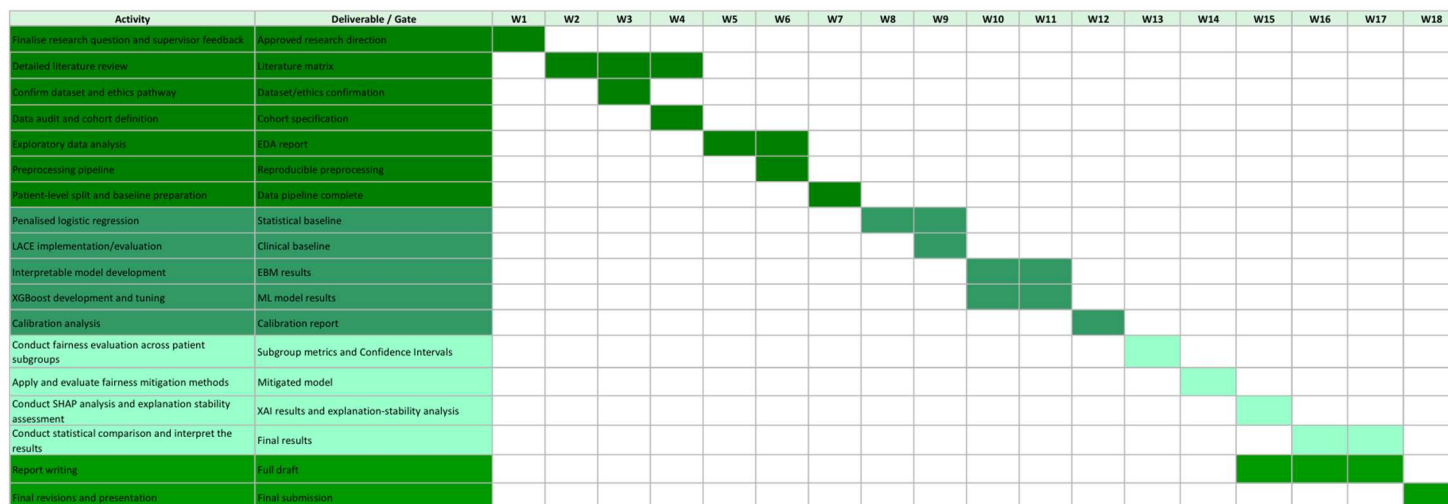
Regarding informed consent, as the data is publicly available and already de-identified, participant recruitment and informed consent are not required for the proposed research. Nevertheless, this research will verify the ethics pathway at Whitecliffe prior to experimental research. The project will not consider the public availability of the data as the absence of ethical concerns.

Regarding algorithmic fairness, it will be one of the key ethical concerns examined in this study. The project will explore whether there are any differences in prediction error rates between races. However, this concept will not be defined in terms of equal probability rates. For instance, if one community has a significantly higher rate of readmission due to unequal healthcare access, enforcing equal probability rates can conceal an existing health discrepancy. Therefore, equality of opportunity in terms of patients experiencing a particular health outcome will be explored by the study, together with calibration and accuracy measures.

Regarding **the Treaty of Waitangi, principles of partnership, participation and protection** will be taken into consideration during the research governance process. As the current project uses a publicly available dataset from the USA and no data on Māori patients will be collected, engaging Māori participants for the purposes of the dataset is unnecessary (Tribunal, 2026). However, this research will recognise the constraints associated with generalising demographic characteristics from a U.S. dataset to New Zealand, avoid making generalisations concerning Māori or Pacific groups, seek guidance from a supervisor and institution about cultural and ethical issues, Consider Māori data sovereignty principles if the future projects would use New Zealand health data.

3.4. Tentative Timeline

Gantt Chart



4. Conclusion

Summary

5 In this proposal, a quantitative study of fairness-aware and interpretable machine learning for prediction of 30-day hospital readmission will be conducted. Whereas other research may emphasize just predictive performance, the present research seeks to look at the relationship between the four core characteristics of responsible artificial intelligence in healthcare: predictive performance, calibration, fairness, and interpretability.

36 The publicly available Diabetes 130-US Hospitals dataset including 101,766 encounters will be utilised in this project. The following methods will be compared in this study: penalised logistic regression (an intrinsically interpretable model) and XGBoost against LACE clinical baseline with the possibility of validation of reconstruction.

Evaluation of discriminatory power will be based on ROC-AUC and PR-AUC metrics, whereas reliability of predicted probabilities will be measured using calibration curves, Brier score and slope/intercept of calibration. For evaluation of fairness, the metric of equal opportunity and differences in true-positive rates between races will be considered. Bootstrap confidence intervals will help to separate the real differences from statistical noise.

Interpretability will be evaluated critically. For explanation of models, the SHAP method will be used. However, explanation quality will be measured in terms of stability, agreement and clinical plausibility.

Next Steps

17 In the next phase of this research, the study cohort will be finalised along with preprocessing of the Diabetes 130-US Hospitals dataset, and subsequently, penalised logistic regression, an inherently interpretable model, and XGBoost will be developed. Wherever necessary variables are available, LACE Index will be determined as a clinical benchmark. These models will be assessed using measures of discrimination and calibration and compared across different age, gender, and racial groups using Equal Opportunity and bootstrap confidence intervals. SHAP-based explanations will be examined to find out important predictive factors and ascertain whether explanations differ across demographic groups. In the end, the concept of fairness-aware threshold adjustment will be explored to see if disparities can be minimised and what trade-offs will emerge from it in terms of predictive ability, calibration, and explanations. The findings will then be interpreted in the light of the research questions, and after that, the final thesis, limitations, ethical concerns, and future research directions will be presented.

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